Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/17/2025
PI Name:	Overbey, Eliah Ph.D.		
Project Title:	High-Resolution, Murine Spatial Transcriptome Mapping of the Impact of Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Dr. Overbey is now an Assistant Professor of Bioastronautics at the University of Texas - Austin. Her email has been updated. Former email at Cornell: elo4004@med.cornell.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	02/01/2021	**End Date:**	01/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/31/2024 per NSSC information (Ed., 4/25/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mason, Christopher Ph.D. (MENTOR: Weill Medical College of Cornell University) Mao, Xiao Wen M.D. (Loma Linda University) Giacomello, Stefania Ph.D. (Kungliga Tekniska Hogskolan (KTH) Royal Institute of Technology, Sweden)		
Grant/Contract No.:	80NSSC21K0316		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Some of the greatest gaps in knowledge regarding the risks of long-duration spaceflight are the effects of spaceflight on the human body and the resulting health consequences. One way to predict these consequences is to observe the cellular and molecular changes that occur in mammalian tissues (e.g., mouse models) and extrapolate their relevance using knowledge of diseases that share similar genomic alterations.
This project originally aimed to use advanced molecular profiling techniques to enhance understanding of spaceflight-induced gene expression changes. Specifically, it proposed to apply single-cell RNA sequencing and spatial transcriptomics to tissues from rodent spaceflight missions archived in the Life Sciences Data Archive (LSDA) and the International Space Station (ISS) National Laboratory. These high-resolution methods would allow for the detection of gene expression changes at the level of individual cell types and provide spatial insights into transcriptional heterogeneity across complex tissues, such as brain and retina. The project was organized into three aims:

Aim 1: Data Generation via single-cell RNA sequencing using the 10x Genomics Chromium system.

Aim 2: Data Generation via spatial transcriptomics using the 10x Genomics Visium system.

Aim 3: Data Analysis to identify differentially expressed genes at the cell-type level and assess spatial patterns of transcription using established software tools and comparison to the NASA GeneLab repository.

While Aims 1 and 2 are completed, Aim 3 (data analysis and interpretation) was not completed during the period of performance and is anticipated to be published this summer (2025) as part of a spaceflight paper collection in Cell Press.

In parallel, the Space Omics and Medical Atlas (SOMA) project on the Inspiration4 (I4) mission, the first all-civilian human spaceflight, was successfully completed. This component of the research provided a complementary human-focused dataset to the rodent omics work. The SOMA-I4 study generated multi-omics datasets (including genomics, epigenomics, transcriptomics, and proteomics) from all four civilian astronauts before and after spaceflight. The datasets represent one of the most comprehensive human spaceflight molecular profiling efforts to date and are being integrated into open-access repositories to support future research on human adaptation to spaceflight.

Collectively, the work performed under this fellowship has contributed to expanding NASA’s understanding of the biological impact of spaceflight on both animal models and humans, while also generating foundational datasets that will inform countermeasure development and risk modeling for future missions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research advances our understanding of how spaceflight affects biological systems at the molecular level, with implications for both astronaut health and medical challenges on Earth. The work includes two distinct components: (1) high-resolution spatial and single-cell transcriptomic profiling of mouse retina tissue from spaceflight missions, and (2) comprehensive multi-omics analysis of the civilian astronauts from the Inspiration4 (I4) mission. Each dataset contributes novel insights with Earth-based health relevance. From the mouse spaceflight retina dataset: - Using single-cell RNA sequencing and spatial transcriptomics on retina tissue from rodents flown in space, we identified cell-type-specific and spatially localized gene expression changes associated with spaceflight. - These changes included markers of oxidative stress, photoreceptor dysregulation, and mitochondrial impairment, aligning with symptoms observed in spaceflight-associated neuro-ocular syndrome (SANS), a condition that affects astronaut vision during long-duration missions. - By mapping transcriptional changes within the spatial architecture of the retina, the study provides a model for understanding the cellular mechanisms that underlie ocular pathophysiology under stress; insights that may inform diagnostics and treatments for retinal diseases such as macular degeneration, diabetic retinopathy, and glaucoma on Earth.

The Inspiration4 (I4) civilian astronaut study, as part of the Space Omics and Medical Atlas (SOMA), generated multi-omics profiles from blood, skin, and microbiome samples from a pool of collected 2,900 specimen aliquots. Key findings include: - Telomere elongation and epigenetic remodeling, offering insight into DNA repair and aging. - Immune activation and viral reactivation signatures, with parallels to autoimmune and inflammatory diseases. - Protein biomarkers of brain, muscle, and kidney stress, informing diagnostics for frailty, neurodegeneration, and sarcopenia. - Sex-specific recovery patterns and inter-individual variability, supporting future personalized medicine strategies.

Technological innovations—particularly protocols for remote sample collection, stabilization, and multi-omics analysis—demonstrate that advanced diagnostics can be deployed outside of clinical settings. During the Inspiration4 mission, astronauts and scientists, including myself, successfully collected and preserved blood, skin biopsies, and microbiome samples using minimal equipment and remote guidance. These methods were validated for downstream genomic, transcriptomic, and proteomic analysis, establishing a model for decentralized precision health monitoring.

Such capabilities have direct relevance on Earth: they enable remote healthcare delivery in rural, underserved, or emergency settings, where access to specialized labs is limited. The ability to self-collect clinically viable samples and ship them for high-resolution analysis opens new possibilities for diagnosing and managing aging-related diseases, immune dysfunction, and chronic conditions in populations with limited access to medical infrastructure.

Over the reporting period, significant progress was made in advancing both human and animal model research on the biological impacts of spaceflight. Work under this fellowship contributed to two major projects: (1) the completion of the Space Omics and Medical Atlas (SOMA) study of the Inspiration4 (I4) civilian spaceflight mission, and (2) progress on Aims 1 and 2 of a high-resolution molecular analysis of mouse retina tissues exposed to spaceflight.
As lead investigator for the SOMA-Inspiration4 project, Dr. Eliah Overbey coordinated the collection and analysis of more than 2,900 biological samples from the I4 crew across multiple time points—before, during, and after spaceflight. This work produced one of the most comprehensive molecular datasets ever assembled from a human spaceflight mission. Samples included blood, skin biopsies, saliva, nasal swabs, stool, and environmental swabs from the spacecraft interior, and were processed for genomic, epigenomic, transcriptomic, proteomic, immune, and microbiome profiling.

The I4 data revealed that even short-duration spaceflight leads to measurable biological changes across nearly all systems. For example, the crew exhibited transient telomere elongation, epigenetic remodeling in immune cells, and altered expression of genes involved in inflammation, DNA repair, and circadian regulation. Protein biomarkers of brain stress, kidney function, and immune activity were

Task Progress:	detected in blood samples post-flight, with many changes returning to baseline within six months. These findings mirror changes seen in longer-duration missions and reinforce the value of short flights for modeling human adaptation to space. In addition to mapping biological responses, the I4 study demonstrated practical innovations for remote health monitoring. Civilian astronauts with no prior medical training successfully collected and preserved biospecimens using lightweight tools and minimal onboard infrastructure. These protocols—developed in collaboration with commercial spaceflight partners—have broad implications for health care on Earth, especially in remote or resource-limited environments. They support the vision of decentralized, high-resolution diagnostics that could be used in telemedicine, emergency response, or rural clinics. Parallel to this work, progress was made on Aims 1 and 2 of the rodent retina study, which uses single-cell and spatial transcriptomic technologies to examine how spaceflight affects vision-related tissues. Retina samples were obtained from archived NASA rodent missions and processed using the 10X Genomics Chromium system (for single-cell RNA sequencing) and Visium system (for spatial transcriptomics). These cutting-edge methods enable the mapping of gene expression changes at the cellular level and within the spatial architecture of the retina. Data from these experiments revealed cell-type-specific transcriptional changes associated with spaceflight, including markers of oxidative stress, mitochondrial dysfunction, and photoreceptor cell remodeling. These findings are directly relevant to understanding spaceflight-associated neuro-ocular syndrome (SANS), a condition that affects vision in a large proportion of astronauts on long-duration missions. By focusing on a highly ordered and complex tissue such as the retina, this work provides key insights into how microgravity may disrupt tissue-specific homeostasis in the eye. Although data generation for both single-cell and spatial assays was completed, Aim 3—the integrated data analysis phase—was not finalized during the reporting period. This step will involve identifying differentially expressed genes across cell types, correlating spatial expression gradients, and comparing results to public omics datasets in the NASA GeneLab repository. This analysis is planned for the next phase of the project, with the goal of generating publishable insights into tissue-level adaptation to microgravity. Together, the SOMA-Inspiration4 study and the retina transcriptomics project represent complementary efforts to understand the molecular consequences of spaceflight in humans and mammalian models. Both projects leverage high-throughput, single-cell, and spatial omics technologies to dissect how environmental stressors—particularly microgravity and radiation—affect gene regulation, tissue function, and immune adaptation. These efforts not only advance NASA's human health risk models for future exploration-class missions but also create valuable data and methods for use in terrestrial medicine. Insights into immune dysregulation, tissue stress, and rapid physiological adaptation have direct relevance for conditions such as inflammation, neurodegeneration, and aging on Earth. Furthermore, the technical innovations in remote biospecimen collection and analysis are paving the way for broader use of precision medicine in non-clinical environments. The progress made in this reporting period sets the stage for impactful publications and further development of translational tools. Analysis of the retina dataset (Aim 3) will continue in the upcoming year, and follow-on studies are in development to expand on the spaceflight-related findings in both human and rodent systems.

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